

A transdiagnostic clinical–biological map of severe mental illness: metabolic and inflammatory load are largely independent of overall functional severity

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Abstract

Psychiatric diagnosis aggregates biologically and prognostically heterogeneous patients into single categories. We reorganized the harmonized three-cohort FACE baseline (bipolar disorder [BP], schizophrenia [SZ], major depression [DR]; $N = 9,013$) around continuous, diagnosis-agnostic axes of variation, estimated by one global, missingness-aware Bayesian sparse bifactor/ESEM model with mixed likelihoods, fit to each patient’s *observed cells only* (no imputation, full sample), with diagnosis held out as metadata. From ten candidate constructs the data resolved a **nine-dimension map**—its continuous backbone certified at full N , the joint mixed fit documented at the largest N that mixes: a general functional-burden factor (G) plus eight weakly correlated axes—cognition, metabolic, inflammatory, sleep, developmental-risk, suicidality, mania and substance. The central finding is that **metabolic and inflammatory load are the least severity-entangled domains** (correlation with G of 0.12 and 0.07, versus 0.39 for cognition and 0.42 for sleep). On these coordinates the patient space is a graded **continuum, not discrete biotypes** (five structure-discovery methods agree; adjusted Rand index ≈ 0 against the DSM-5 subtypes), describable by eight extreme phenotypes whose two-year functional-remission rates range 14%–60%. Scored forward onto follow-up the geometry *persists*—durable biology, moving symptoms. It also *predicts* future functioning incrementally beyond diagnosis and severity, but modestly and at the group level (functional-remission AUC +0.017; course-dependent). On observational treatment-as-usual the map does *not* reliably moderate response. FACE-ATLAS provides a real, stable, transdiagnostic clinical–biological continuum and a deliberately calibrated account of its prognostic reach; establishing individual-level utility or treatment guidance would require incident-event outcomes, randomized treatment data and external validation.

Keywords: transdiagnostic psychiatry; dimensional psychopathology; bifactor model; metabolic syndrome; inflammation; bipolar disorder; schizophrenia; major depression; normative modelling; precision psychiatry.

1 Introduction

Psychiatric diagnoses are clinically indispensable but biologically porous. The DSM categories aggregate patients who differ widely in their biology, course and treatment response, and they are undercut by pervasive comorbidity, within-category heterogeneity and diagnostic instability over time. In response, two influential programmes have argued that psychopathology is better described by continuous, *transdiagnostic* dimensions than by categorical boundaries: the U.S. National Institute of Mental Health’s Research Domain Criteria (RDoC) [1] and the Hierarchical Taxonomy of Psychopathology (HiTOP) [2, 3]. A recurring empirical motif within this work is a

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Table 1: **Sample characteristics by cohort (FACE baseline, $N = 9,013$).** Aggregate-only summary; sex coded female. Per-patient data are confidential (Fondation FondaMental). Retention rows are the percentage of the baseline cohort re-assessed at the 12- and 24-month visits.

Characteristic	Bipolar	Schizophrenia	Depression	Overall
N	6,252	2,209	552	9,013
Age, years — mean (SD)	39.3 (13.4)	31.1 (9.6)	51.0 (13.3)	38.0 (13.4)
Age, years — median [IQR]	38 [28–49]	29 [23–37]	52 [42–60]	36 [27–48]
Female — n (%)	3,969 (63.5)	600 (27.2)	323 (58.5)	4,892 (54.3)
Education, years — mean (SD)	14.2 (2.8)	12.3 (2.6)	13.2 (3.2)	14.0 (2.8)
Recruitment sites — n	15	12	13	21
Education missing — n (%)	1,803 (28.8)	1,646 (74.5)	228 (41.3)	3,677 (40.8)
<i>DSM-5 subtype — n (% of cohort)</i>				
Bipolar I	2,635 (42.1)	—	—	
Bipolar II	2,956 (47.3)	—	—	
Bipolar, unspecified	661 (10.6)	—	—	
Schizophrenia	—	1,692 (76.6)	—	
Schizoaffective disorder	—	476 (21.5)	—	
Schizophreniform disorder	—	41 (1.9)	—	
Major depressive disorder	—	—	552 (100.0)	
Retained at 12 months — %	49.2	43.6	42.2	47.4
Retained at 24 months — %	35.6	26.9	24.5	32.8

single general factor that spans disorders—the “ p factor”—most defensibly read as a dimension of life impairment rather than a latent liability to specific symptoms [4, 5], although the substantive meaning of such general factors remains contested [6–8].

This dimensional structure has, however, been mapped almost entirely from a single kind of measurement. Symptom-based work [2, 4] and brain-imaging work [9–11] each describe one slice of the patient. Routinely collected *biology*—the metabolic and inflammatory markers that dominate the physical-health burden of severe mental illness—has not been placed *inside* the transdiagnostic factor space across disorders, even though an independent literature suggests it forms a partly separable axis: metabolic and inflammatory abnormality co-occur in roughly a fifth to a third of mood-disorder patients and map onto atypical, energy-related symptoms rather than overall severity [12–15], and metabolic burden in severe mental illness tracks medication, adiposity and chronicity more than symptom load [16–19]. Whether biological load is genuinely decoupled from a transdiagnostic severity axis has never, to our knowledge, been measured directly.

A second question is structural. If patients vary continuously, the actionable object is position on an axis; if they fall into discrete biological kinds, it is group membership. Normative-modelling studies of brain structure in the disorders studied here report heterogeneity so extensive that “the average patient” is barely informative [9–11]. Against this, the B-SNIP programme has identified replicable, brain-based psychosis “biotypes” that cut across DSM boundaries [20–22]. The continuum-versus-biotype question is therefore genuinely open, and may depend on which measurement space is interrogated [23].

Here we address both questions in one analysis. Using the harmonized baseline of the three FACE cohorts—bipolar disorder, schizophrenia and major depression, $N = 9,013$ —we estimate a single global, missingness-aware Bayesian sparse bifactor/ESEM model with mixed likelihoods, fit to each patient’s observed cells only, that places clinical, cognitive, behavioural and laboratory measurements in one coordinate system while holding diagnosis out as metadata. We then ask, in five dependent steps, how far such a map can be pushed: does it *exist* (a stable dimensional structure), does it *organize* patients (into biotypes or a continuum), does it *persist* over two years

Table 2: **The nine-dimension transdiagnostic map: anchoring indicators, strength, severity entanglement, adjudication verdict and measurement invariance.** The continuous backbone is certified at full N ; the joint mixed fit is documented at the largest N that mixes. Mean primary $|\lambda|$ is the average home loading; correlation with G is from the correlated- G model (continuous backbone); for mania and substance (outside the continuous backbone) the entry is the bifactor direct $|\lambda_G|$. Verdicts are outcomes of the analysis. Depression/anxiety instruments are cross-loading windows on G (loadings 0.66–0.80), not a separate factor; anhedonia was rejected; impulsivity, negative symptoms and sensory abnormality were not testable for lack of common indicators.

Dimension	Block / likelihood	Anchoring indicators	Mean $ \lambda $	corr G	Verdict	Invariance
G (functional burden)	explicit, Gaussian	FAST 0.90, EGF, EQ-5D; no symptom content	—	—	confirmed (severity)	partial (SZ: no FAST)
Cognition	marginalized	TMT, WAIS coding/digit span, fluency	0.57	0.39	confirmed	invariant
Metabolic	marginalized	BMI, lipids, glycaemia, blood pressure	0.32	0.12	confirmed (split)	invariant
Inflammatory	marginalized	CRP, leukocyte subsets	0.39	0.07	confirmed (split)	non-invariant (DR: φ 0.71/0.75)
Sleep	marginalized	PSQI total + subscales	0.48	0.42	confirmed	most invariant (φ 0.99)
Developmental-risk	explicit	childhood adversity (CTQ), perinatal, family history	0.42	—	confirmed (proxy)	invariant
Suicidality	explicit, binary	ISF ideation/attempt (+2.2–2.7 logit)	strong	—	confirmed (mixed)	3-cohort ISF
Mania	marginalized	YMRS, Altman	0.49–0.76	0.15	confirmed	partial (DR: Altman floors)
Substance	explicit, mixed	alcohol/cannabis SUD, nicotine (+0.38–0.69 logit)	—	0.13	confirmed	invariant BP–SZ (2-cohort)

of follow-up, does it *predict* clinical outcomes beyond diagnosis and severity, and does it *guide treatment*? The answers are, respectively: a nine-dimension map on which biology is the least severity-entangled domain; a graded continuum rather than discrete biotypes; a geometry that persists, with durable biology and moving symptoms; a modest, group-level prognostic signal for functioning; and—on observational treatment-as-usual—no reliable treatment moderation. We report the modest and null results as deliberately as the positive ones: the contribution is a calibrated map, not an overstated one.

2 Results

2.1 Sample and analytic strategy

The analysis sample was the harmonized baseline (visit V0) of the three FACE cohorts: $N = 9,013$ patients (bipolar disorder [BP] 6,252; schizophrenia [SZ] 2,209; major depression [DR] 552) recruited across 21 expert-centre sites (Table 1). The cohorts differ in age, sex distribution and chronicity, as expected for these disorders. We deliberately fit the measurement model on the *full* sample without completeness selection; 143 indicators were modelled, with a mean cell missingness of 39.8% preserved as missing and never imputed. Diagnosis (cohort and DSM-5 subtype) was held out of the model and used only as a validation/metadata label. Follow-up visits at 12 and 24 months (retained: 47.4% and 32.8% of V0) were used only to test temporal coherence and prognosis, never to define the map. The full statistical specification is in Methods; Figure 1 summarizes the design.

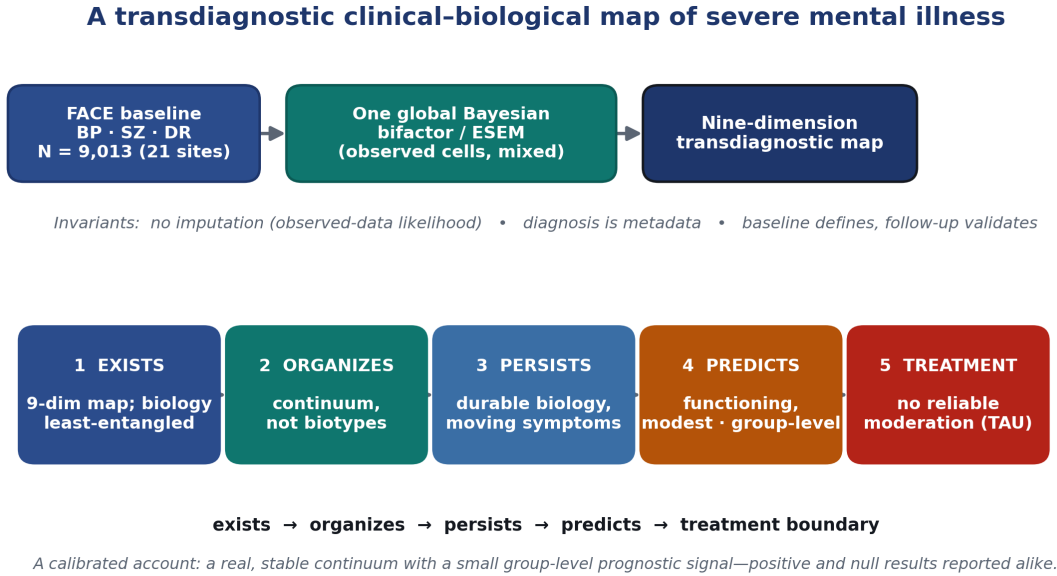


Figure 1: **Study design and the five questions.** The harmonized three-cohort FACE baseline (bipolar disorder, schizophrenia, depression; $N = 9,013$) is reorganized around diagnosis-agnostic axes by one global, missingness-aware Bayesian bifactor/ESEM model fit to each patient’s observed cells only, with three load-bearing invariants (no imputation, diagnosis as metadata, baseline defines/follow-up validates). The fitted map is then interrogated through five dependent questions—exists, organizes, persists, predicts, guides treatment—with positive and null results reported alike.

2.2 A nine-dimension transdiagnostic map (the map exists)

From ten candidate constructs, one global model returned a stable nine-dimension structure: a general functional-burden factor (G) plus eight weakly correlated specific axes—cognition, metabolic, inflammatory, sleep, developmental-risk, suicidality, mania and substance (Figure 2; Table 2). Inter-factor correlations were weak (mean absolute off-diagonal ≈ 0.10), confirming genuinely distinct axes rather than one collapsed dimension. G was anchored only by functioning and global-severity indicators (FAST loading 0.90, EGF 0.73) and carried no symptom content, supporting its conservative reading as transdiagnostic functional burden rather than a symptom-liability “ p ”. Depression and anxiety instruments (MADRS, QIDS, STAI) did not form a separate factor; they loaded 0.66–0.80 on G as cross-loading “windows”. The single prior “biology” candidate *split* into distinct metabolic and inflammatory factors (inter-factor $\Phi \approx 0.19$). Of the ten candidates, nine were confirmed (one as a split), anhedonia was rejected, and impulsivity, negative symptoms and sensory abnormality were *not testable* for lack of common indicators—each verdict an outcome of the analysis rather than an assumption (Table 2). The structure was not an artefact of the priors: a flat-prior refit reproduced the loadings and Φ essentially exactly (Tucker congruence $\varphi = 1.00$), and model comparison decisively preferred the bifactor over unidimensional and correlated-factor alternatives. Absolute fit was acceptable across both likelihood blocks (continuous Bayesian SRMR ≈ 0.07 ; 21/22 non-Gaussian indicators reproduced within the 90% posterior-predictive interval).

2.3 Biology is the least severity-entangled domain (the headline)

Because the bifactor identification fixes $G \perp$ specifics by construction, we re-estimated the model allowing G to correlate freely with the specific axes. Under this correlated- G model, metabolic and inflammatory load were the *least* entangled with functional burden: correlations with G of 0.12 (metabolic) and 0.07 (inflammatory), against 0.39 for cognition and 0.42 for sleep (Figure 3). In other words, two patients with the same overall impairment can carry very different metabolic and immune loads—a patient’s biological burden rides largely on axes that the global clinical-severity picture does not capture. Consistent with a large independent literature (see Discussion), we state this as relative, not absolute, independence: biology is *largely independent of a general functional-burden axis*, with the 0.07–0.12 values quantifying—at the low end of what that literature would predict—a pattern previously described only qualitatively.

2.4 The patient space is a continuum, not biotypes (the map organizes)

Before clustering, a pre-registered structure-discovery gate—run over the model’s posterior draws and combining cluster tendency, modality, density, model selection and topology—returned a unanimous verdict of *no discrete clusters*: gap-statistic $K = 1$; HDBSCAN assigned 100% of points to noise; the first principal component was unimodal (Hartigan dip $p \approx 0.99$); the archetype scree showed no elbow; and the mixture BIC formed a flat basin. The nine-dimensional space is therefore a graded continuum. We represented it two complementary ways: eight extreme phenotypes (archetypes; cross-seed congruence 0.999), each the high pole of one axis plus a low-burden pole, with every patient a soft convex blend (Figure 4); and a coarser four-region soft tessellation. Three properties make the continuum decision-relevant. First, it is *transdiagnostic*: the adjusted Rand index between the data-driven structure and diagnosis was ≈ 0 (tessellation 0.02 vs the seven DSM-5 subtypes; archetypes 0.05), every phenotype mixing all cohorts. Second, it is *driven by the specific/biological axes, not severity*: partition variance explained peaked on specific axes (maximum $\eta^2 = 0.45$) above G (0.31), and—crucially—distinct metabolic and inflammatory corners emerged, so the biology $\perp G$ signal survives into the phenotypes, while overall severity formed no corner of its own (it is the continuum’s spine). Third, it is *not a missingness artefact*: a classifier predicting membership from the per-axis coverage pattern

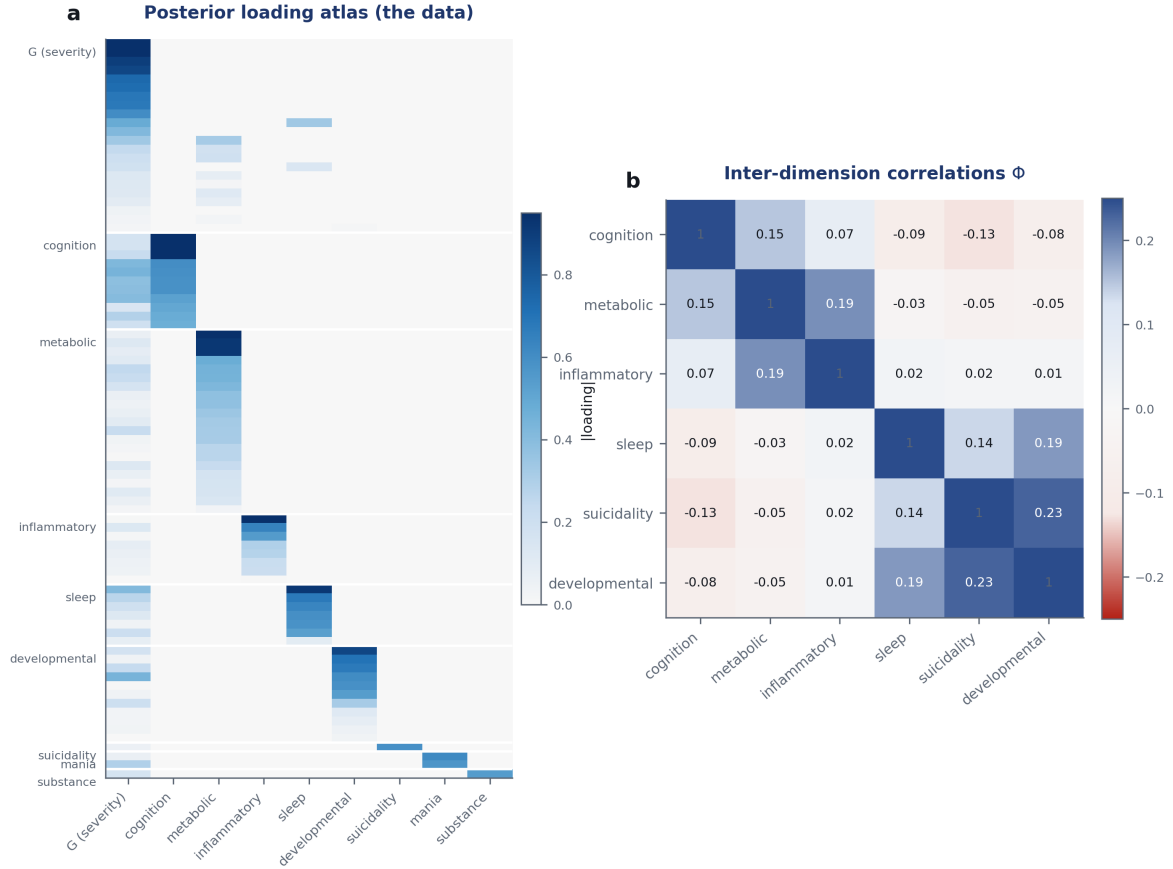


Figure 2: **The nine-dimension map.** (a) Posterior loading atlas (indicators $\times$ factors); colour encodes the absolute posterior loading, with rows grouped by home factor. Each specific axis is anchored by a clinically coherent indicator block, while G (functional burden) carries the cross-loading depression/anxiety windows. (b) The inter-factor correlation matrix Φ among the specific axes is weak throughout (mean $|\text{off-diagonal}| \approx 0.10$), confirming distinct axes; metabolic and inflammatory correlate only ≈ 0.19 .

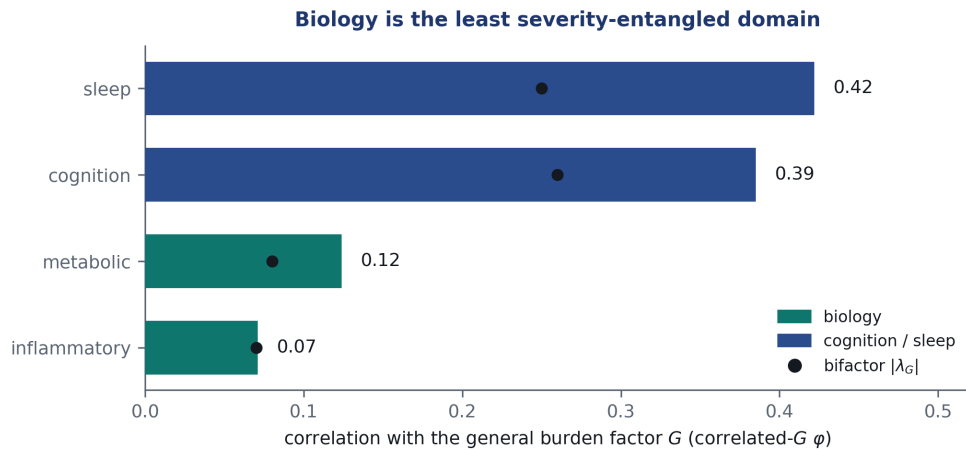


Figure 3: **Biology is the least severity-entangled domain.** Correlation of each specific axis with the general functional-burden factor G under the correlated- G model (bars), with the bifactor general loading $|\lambda_G|$ (points) shown for comparison. Metabolic and inflammatory load are the least severity-entangled domains, whereas cognition and sleep partly track G .

performed below baseline (lift -0.08), and the partition was seed-stable. The data-driven regions also described the coordinate cloud more tightly than diagnosis (mixture BIC 199,325 vs 206,016 for a DSM-5-constrained model; coordinate variance explained $\approx 21\%$ vs $\approx 5\%$)—a descriptive, not yet clinical, superiority.

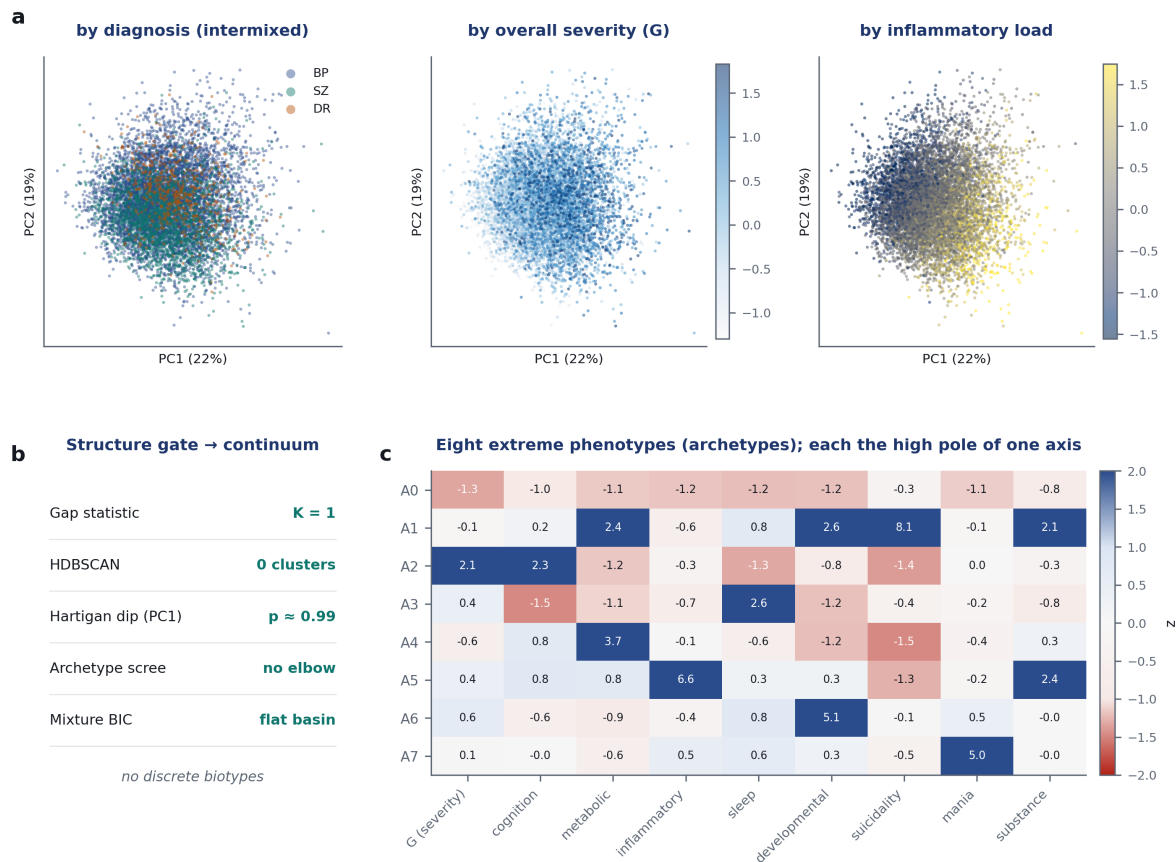


Figure 4: **The patient space is a continuum, not biotypes.** (a) Principal-component embedding of the nine-dimensional coordinates ($N = 9,013$), coloured by diagnosis (fully intermixed), by overall severity G (a smooth gradient along PC1), and by inflammatory load (a gradient in a different direction—biology $\perp G$). (b) Five structure-discovery methods agree on no discrete clusters. (c) Standardized profile of the eight extreme phenotypes (archetypes) across the nine axes; each is the high pole of one axis (note distinct metabolic and inflammatory corners), with most patients soft blends of several.

2.5 The geometry persists over two years (the map persists)

Scored forward onto the 12- and 24-month visits on the *fixed* model (observed cells, uncertainty propagated, never re-estimated), the measurement remained invariant for five of six testable backbone axes (Tucker $\varphi = 0.96$ – 1.00 ; inflammatory partial at 0.90), licensing the reading of coordinate change as patient change rather than instrument drift. A measurement-error decomposition then separated trait from state (Figure 5). The durable, trait-like axes were the biological and cognitive ones (rank-stability intraclass correlation: metabolic 0.93 , inflammatory 0.85 , cognition 0.78); the moving, state-like axes were the symptom ones (sleep 0.49 , suicidality 0.46). At the population level the cohort slid *down* the severity and symptom axes while remaining static on biology. A per-patient geometric analysis agreed: severity moved (cleared measurement error) in 34.5% of patients versus only 20.2% for the biology corner; the “spine slides while biology holds” pattern (25.8%) outnumbered its converse (11.5%) by $2.2\times$. Archetype

identity persisted in 52% of patients ($\kappa = 0.27$ vs a 12.5% chance rate). The clinical reading is compact: *stratify on the durable biology, monitor the moving symptoms*.

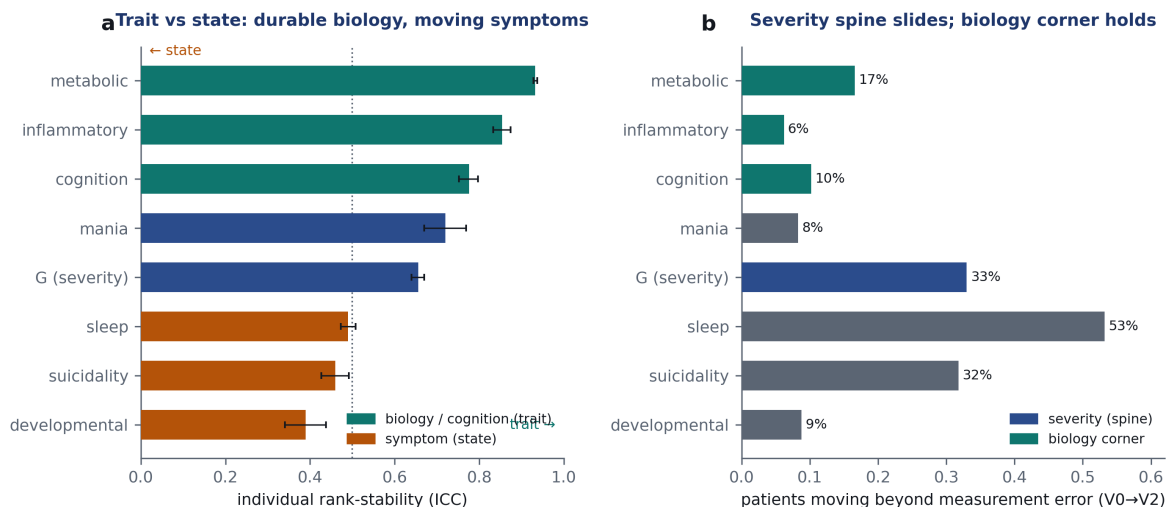


Figure 5: **Durable biology, moving symptoms.** (a) Individual rank-stability (intraclass correlation) over follow-up: the biological and cognitive axes are trait-like (high), the symptom axes state-like (low). (b) Proportion of patients whose position changes beyond measurement error from baseline to 24 months: the severity spine moves far more than the biology corner.

2.6 It predicts future functioning, modestly and at the group level (the map predicts)

We then asked whether a baseline coordinate or phenotype predicts a two-year outcome *incrementally beyond* diagnosis, current severity and the baseline value of the outcome, using an errors-in-variables Bayesian model that propagated each coordinate’s measurement uncertainty. The map added predictive signal for *functioning* but not for severity (which was autoregression-saturated). The eight archetypes added held-out predictive value for functioning (expected log-predictive-density gain +46), and the durable biological axes were individually informative: higher metabolic ($\beta = -0.062$, 95% credible interval $[-0.103, -0.022]$) and inflammatory ($\beta = -0.060$ $[-0.112, -0.011]$) load predicted worse future functioning, with the metabolic effect surviving adjustment for the error-corrected severity coordinate—it is not baseline severity measured noisily. The signal was *co-informative* with DSM-5 (each added beyond the other) and *course-dependent*, concentrated in the episodic courses (bipolar disorder, depression) and null in the more baseline-saturated schizophrenia presentations. Translated to the clinic, the eight archetypes sorted patients into groups whose two-year functional-remission rate ranged from 14% to 60%, transdiagnostically (Figure 6). The individual-level discrimination gain was nonetheless small (functional-remission AUC +0.017 $[+0.009, +0.026]$ over a strong diagnosis-plus-severity reference): the map’s value is group-level prognostic stratification and continuous functional forecasting, not a large boost to individual binary prediction. These results survived inverse-probability weighting for attrition, restriction to reliably measured axes, and a permutation null.

2.7 It does not yet guide treatment—a boundary drawn by the evidence

Finally, treatment data recovered from the per-cohort source dictionaries and harmonized to drug-class exposures were run through a causal pipeline (common-support overlap gate $\rightarrow$ propensity weighting $\rightarrow$ doubly-robust errors-in-variables moderation with E-value sensitivity analysis) to ask whether the durable axes *moderate* treatment response. On observational treatment-as-usual

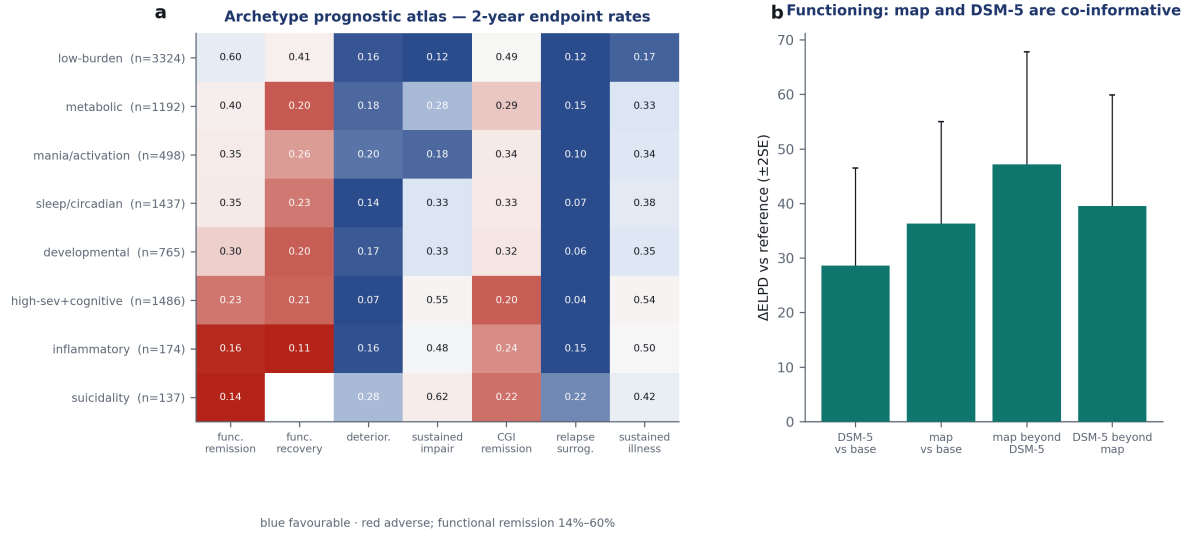


Figure 6: Prognostic reach for functioning. (a) Archetype prognostic atlas: two-year clinical-endpoint rates by baseline archetype (blue favourable, red adverse); functional-remission rates span 14%–60%, every archetype transdiagnostic. (b) Held-out incremental value for functioning (EGF): the map adds beyond a diagnosis-plus-severity foundation *and* DSM-5 adds beyond the map—the two are co-informative (mean ± 2 SE).

they did not reliably do so. Where identification was cleanest (lithium in bipolar disorder, near-complete overlap) the result was a well-identified null; an antipsychotic $\times$ metabolic/inflammatory signal for functioning in bipolar disorder was suggestive but unconfirmed (average-effect E-value 1.79); and clozapine in schizophrenia was channelled to treatment-resistant patients and hence non-estimable. Average treatment effects were uniformly confounding-fragile (E-values 1.1–1.8), the expected signature of confounding by indication (Extended Data Fig. 7). Importantly, the prognostic finding above was not merely unmodelled treatment: the metabolic $\rightarrow$ functioning forecast survived adjustment for the drug classes patients were actually receiving (attenuation 4.4%, credible interval still excluding zero). The boundary is therefore earned, not assumed: demonstrating treatment *selection* from this map would require randomized or trial-arm data, which this baseline observational cohort lacks.

3 Discussion

Across three cohorts of severe mental illness, a single missingness-aware Bayesian model placed clinical and biological measurements in one transdiagnostic coordinate system and yielded a nine-dimension map whose most consequential property is that metabolic and inflammatory load are the domains least entangled with overall functional severity. If biological burden simply tracked how impaired a patient is, it would add little to a clinician’s global impression. The data say the opposite: biological risk rides on axes that the severity picture does not see, which is the precise property a biology-aware stratification can exploit—separating patients who look equally ill but are biologically opposite.

This headline is corroborated, in direction, by a large and independent literature, which is why we state it as relative rather than absolute independence. Metabolic burden in severe mental illness is driven by medication, adiposity, age and chronicity more than by diagnosis or symptom severity, is present before chronic severity accrues, and is comparable across schizophrenia, bipolar disorder and depression [16–19, 24]; inflammation marks a subgroup and specific symptoms rather than the severity continuum [25–27]; and the immuno-metabolic-depression programme has

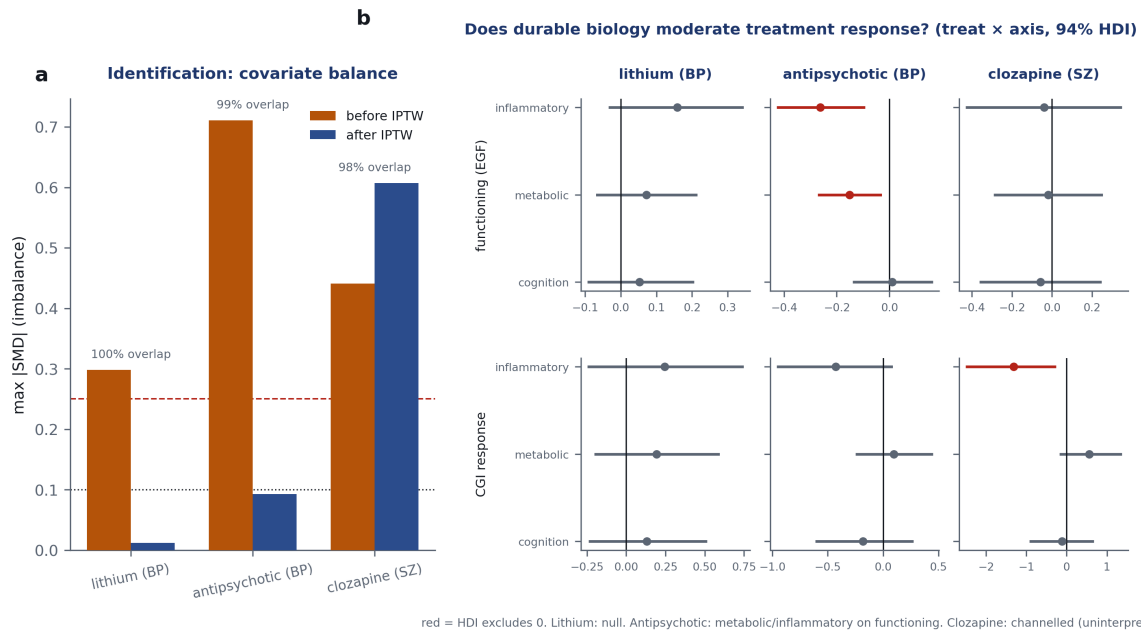


Figure 7: Extended Data Fig. 1 | Treatment moderation is not reliably identified on observational treatment-as-usual. (a) Identification gate: maximum covariate imbalance (standardized mean difference) before and after stabilized inverse-probability-of-treatment weighting. Lithium in bipolar disorder balances to ≈ 0 (near-complete overlap) and antipsychotic balances, but clozapine in schizophrenia cannot be balanced—the signature of channelling. (b) Doubly-robust durable-axis $\times$ treatment moderation (94% highest-density intervals) for functioning and CGI response; red intervals exclude zero. Lithium is a well-identified null; the antipsychotic metabolic/inflammatory $\times$ functioning signal is suggestive but unconfirmed; clozapine is channelled and uninterpretable.

independently arrived at a separable biological dimension mapping onto atypical, energy-related symptoms [12–15, 28]. That a completely different method converges on a separable biology axis is strong convergent validity. We also state the boundary conditions plainly: case–control inflammatory elevations are real [29–31], a modest severity/duration dose–response for metabolic burden exists [32, 33], metabolic dysfunction does track cognition and functioning in mood disorders [34], and much of the inflammation–symptom association is carried by adiposity rather than inflammation *per se* [35, 36]. Our 0.07–0.12 correlations are best read as a quantification of a known qualitative pattern, sitting at its low end because G indexes *functional burden*, from which biology is most decoupled, rather than the case–control contrast of being ill.

The second contribution is structural. On these clinical–biological coordinates the patient space is a graded continuum, not a set of discrete biotypes, and it cuts almost entirely across DSM-5. This aligns with normative-modelling work reporting extensive individual heterogeneity in the same disorders [9–11], and it stands in apparent tension with the brain-based B-SNIP biotypes [20, 21]. The tension is more apparent than real: the two programmes interrogate different measurement spaces, B-SNIP itself found DSM diagnoses to be a severity continuum, and recent syntheses endorse both categorical and dimensional views [22]; clustering choices can also resurface biotypes from continuous data [23]. We therefore claim no biotypes *in this nine-dimensional clinical–biological space*, not that biotypes are absent from all spaces. Anchoring G on functioning alone also sidesteps the well-known instability in what a symptom-only general factor means [6–8], while remaining consistent with the impairment reading of the p factor [4, 5].

The remaining contributions are deliberately calibrated. The map persists over two years—durable biology, moving symptoms—which both validates the measurement longitudinally and motivates a clinical division of labour between traits to stratify on and states to monitor. It predicts future functioning incrementally beyond diagnosis and severity, but the gain is modest and group-level, and it is co-informative with rather than superior to DSM-5; the value lies in prognostic enrichment and continuous forecasting, not individual risk calculation. And on observational treatment-as-usual the map does not reliably moderate response: where identification is clean the result is a well-identified null, and where it is not (channelled prescribing) the effect is non-estimable. We regard reporting these modest and null findings as a feature. The field has repeatedly produced biotype and biomarker claims that fail to replicate; a transdiagnostic biology map that is honest about the difference between scientific validity (demonstrated here) and individual-level clinical utility (not demonstrated here) is a useful corrective.

In sum, FACE-ATLAS reframes a clinically actionable hypothesis with measured coordinates: not “which biotype is this patient?” but “where does this patient sit on continuous, partly biological axes that diagnosis and severity do not capture, and how durable is that position?” The biological axes that a patient keeps over time forecast the functional state they move toward—a group-level, transdiagnostic signal that complements diagnosis and points to a concrete, testable metabolic/inflammatory $\times$ treatment hypothesis for prospective study.

Limitations

Four classes of limitation bound these claims. *Internal validity only*: the map is discovered and validated within FACE, with no external-cohort replication, and several invariance partials are documented (G re-anchors in schizophrenia, which lacks the FAST functioning anchor; the inflammatory axis re-weights in the depression cohort; self-rated mania floors in depression). *Outcomes are scale trajectories, not recorded events*: prognostic endpoints are state transitions defined from repeated functioning and severity scales, not registered relapses or hospitalizations—a de-scope from event-level prognosis. *Observational treatment*: confounding by indication is the dominant threat (E-values 1.1–1.8), exposures are drug-class rather than protocol, and channelled treatments are non-estimable; the treatment analysis can therefore only *bound* moderation, not establish selection. *Measurement coverage*: two axes are thin (mania has two indicators; substance is a two-cohort axis), education is 40.8% missing overall (and 74.5% in

schizophrenia)—a structured missingness the observed-likelihood design handles but that limits that covariate—attrition is informative (addressed by inverse-probability weighting), and the follow-up window is two years across three visits. Establishing individual-level clinical utility or treatment guidance would require incident-event outcomes, randomized or trial-arm treatment data, and external validation beyond this baseline observational cohort.

4 Methods

4.1 Cohorts, harmonization and the no-imputation data layer

Data were the baseline (visit V0) of three FACE cohorts (Fondation FondaMental expert centres): bipolar disorder, schizophrenia and major depression ($N = 9,013$; BP 6,252, SZ 2,209, DR 552; 21 recruitment sites). A harmonized data dictionary maps each modelled variable to its per-cohort source column, a harmonization rule, and per-variable plausibility (“sanity”) bounds; out-of-range values are set to missing (never clipped, never imputed). Deterministic skip-logic decoding sets structural zeros only where a gate is explicitly negative (e.g. attempt count = 0 when ever-attempted = 0), logged with provenance; this is not imputation. Each variable carries a likelihood family and a missingness type. Identifiers (patient, cohort, DSM-5 subtype, visit, site) are never modelled as indicators; cohort and subtype enter only as covariates, invariance-grouping variables, or validation labels. Of 201 usable common variables, 143 entered as modelled indicators across the candidate factors; the remainder are covariates or identifiers.

4.2 Candidate ontology and soft priors

Ten candidate constructs (impulsivity, cognitive flexibility, negative symptoms, anhedonia, metabolism/immunometabolism, sleep/circadian, overall severity, sensory abnormality, neurodevelopment, suicidality)—later joined by mania and substance once their indicators were ingested—define a *soft-prior* ontology rather than a fixed output. Mapping the candidates against the FACE common-variable coverage fixes the eligible set, and is itself a primary result: a construct with no common indicator is reported *not testable*, never back-filled with an invented proxy. Each (indicator, factor) cell is assigned a prior *role*—primary, plausible-cross, unlikely or forbidden—whose formal prior is given in Eq. (12) below; the default is shrinkage, not exclusion, so the soft-prior exploratory structural-equation model (ESEM) [37, 38] can confirm, split, merge, downgrade (proxy), reject or declare not-testable any candidate.

4.3 Generative model and local independence

Index patients $i = 1, \dots, N$, indicators $j = 1, \dots, J$, specific factors $k = 1, \dots, K$ and one general factor G . We posit that a few latent causes generate the many measured quantities: each patient carries a factor vector $f_i = (G_i, D_{i1}, \dots, D_{iK})^\top \in \mathbb{R}^{K+1}$ with

$$f_i \sim \mathcal{N}_{K+1}(\mathbf{0}, \Phi), \quad \text{diag}(\Phi) = \mathbf{1}, \quad (1)$$

where the factor *correlation* matrix Φ has unit diagonal (fixing the latent scale). Each indicator is a linear reflection of the factors plus item-specific noise, through a single linear predictor

$$\eta_{ij} = \alpha_j + \lambda_{jG} G_i + \sum_{k=1}^K \lambda_{jk} D_{ik} + \beta_j^\top c_i, \quad (2)$$

with item intercept α_j , general loading λ_{jG} , specific loadings λ_{jk} (row j of the loading matrix $\Lambda = [\lambda_G \mid \Lambda_D] \in \mathbb{R}^{J \times (K+1)}$), and item-level covariates c_i that adjust an indicator’s *mean* but are never factor indicators. The reported (primary) fit z-scores each indicator and omits the covariate term; covariate adjustment (age, sex, education, site; the $\beta_j^\top c_i / Bc_i$ term, implemented

by residualizing each indicator on those covariates before the factor model, Frisch–Waugh–Lovell-equivalent for Gaussian items) is a *sensitivity* arm, and the biology– G result is reported under both — the adjustment *strengthens* it (metabolic– G $0.12 \rightarrow 0.06$; SI). Stacking one patient’s items, with the general column separated from the specific block,

$$x_i = \mu_i + \underbrace{\lambda_G G_i}_{\text{general}} + \underbrace{\Lambda_D D_i}_{\text{specifics}} + \varepsilon_i, \quad \mu_i = \alpha + Bc_i, \quad \varepsilon_i \sim \mathcal{N}(\mathbf{0}, \Psi), \quad (3)$$

with diagonal residual covariance $\Psi = \text{diag}(\sigma_1^2, \dots, \sigma_J^2)$.

Local independence (the defining assumption). Given a patient’s true latent scores, the items are conditionally independent,

$$p(x_i | f_i, \Theta) = \prod_{j \in O_i} p(x_{ij} | f_i, \Theta), \quad (4)$$

i.e. the factors are the only reason items covary. In the Gaussian linear model this is *identical* to “ Ψ is diagonal”: local independence and diagonal residual covariance are two faces of one assumption—that the factors explain away all shared covariance. (It is also why filling missing cells is forbidden: an invented value injects covariance the model would mistake for a factor.) Because f_i is unobserved and the likelihood must depend on Θ alone, we integrate it out by the sum rule and multiply across independent patients,

$$L(\Theta) = \prod_{i=1}^N p(x_i | \Theta) = \prod_{i=1}^N \int \left[\prod_{j \in O_i} p(x_{ij} | f_i, \Theta) \right] \mathcal{N}(f_i; \mathbf{0}, \Phi) df_i. \quad (5)$$

This is the complete first-principles likelihood—a product over patients (independent units), a product over items (local independence), and an integral that removes the latent (the sum rule); everything that follows is the evaluation of (5).

Bifactor / ESEM. Equation (2) is a bifactor exploratory structural-equation model [37, 38]: *bifactor* because every indicator may load on G *and* on its home specific factor; *ESEM* because an indicator is additionally *allowed* small cross-loadings on its plausibly-related factors, shrunk toward zero by the soft priors (12)—unlike confirmatory factor analysis, which forbids cross-loadings outright. The reported fit is a *Bayesian sparse bifactor with selected ESEM windows* rather than a fully free rotation: cells marked UNLIKELY are fixed at (equivalently, shrunk to) zero and specific $\leftrightarrow$ specific cross-loadings are not freed (they are rotationally aliased with Φ); a soft-zero sensitivity that instantiates the $\mathcal{N}(0, 0.05^2)$ prior on the unlikely cells reproduces the map to loading congruence 0.97 (SI). Two identifications of the factor covariance are estimated. In the *bifactor* model (primary) $G \perp D_k$ and $D_k \perp D_l$, so Φ is block-diagonal, $\Phi = \text{blockdiag}(1, \Phi_D)$; this is the cleanest first fit. In the *correlated- G* model (sensitivity) the general–specific block is freed (Eq. (15)), which is what tests whether “ $G \perp \text{biology}$ ” is a finding or an artefact of the bifactor constraint.

4.4 Mixed-likelihood measurement and the observed-data likelihood

Rather than coerce variables onto a common scale—which would manufacture covariance and bias the very factors being estimated—each indicator keeps the observation density appropriate

to its type, all sharing the *same* linear predictor (2):

$$\text{continuous / z-score: } X_{ij} \sim \mathcal{N}(\eta_{ij}, \sigma_j^2), \quad (6)$$

$$\text{skewed laboratory: } \log X_{ij} \sim \mathcal{N}(\eta_{ij}, \sigma_j^2), \quad (7)$$

$$\text{ordinal } (C_j \text{ levels}): \Pr(X_{ij} \leq c) = \text{logit}^{-1}(\tau_{jc} - \eta_{ij}), \quad \tau_{j1} < \dots < \tau_{j,C_j-1}, \quad (8)$$

$$\text{binary: } X_{ij} \sim \text{Bernoulli}(\text{logit}^{-1}(\eta_{ij})), \quad (9)$$

$$\text{count: } X_{ij} \sim \text{NegBinom}(\mu_{ij} = e^{\eta_{ij}}, \phi_j). \quad (10)$$

The continuous block is Gaussian (a Student- t likelihood is available as a robustness option but is not analytically marginalizable, §4.7); the non-Gaussian families fix the factor scale through their cutpoints/offset for identification. Writing $F_j(\cdot)$ for indicator j 's density and $O_i \subseteq \{1, \dots, J\}$ for the *observed* indicators of patient i , the observed-data log-likelihood sums over observed cells only,

$$\log p(X_{\text{obs}} \mid \Theta) = \sum_{i=1}^N \sum_{j \in O_i} \log F_j(X_{ij} \mid \eta_{ij}, \theta_j), \quad (11)$$

so a missing cell contributes no term and no completed $N \times J$ matrix is ever formed; deterministic skip-logic decodes structural zeros as *observed* values (not imputation).

4.5 Soft-prior loadings, identification and the rotation problem

Theory enters through the prior on each loading. With a role $\rho(j, k) \in \{\text{PRIMARY}, \text{CROSS}, \text{UNLIKELY}, \text{FORBIDDEN}\}$ per (indicator, factor) cell,

$$\lambda_{jk} \mid \rho(j, k) \sim \begin{cases} \mathcal{N}_+(0.70, 0.25^2) & \text{PRIMARY (HOME CELL),} \\ \mathcal{N}(0, 0.25^2) & \text{PLAUSIBLE CROSS,} \\ \mathcal{N}(0, 0.05^2) & \text{UNLIKELY,} \\ \delta_0 & \text{FORBIDDEN,} \end{cases} \quad (12)$$

where $\mathcal{N}_+$ truncates to $(0, \infty)$ and δ_0 fixes the loading at zero. Remaining priors are weakly informative: $\alpha_j \sim \mathcal{N}(0, 1.5^2)$; $\sigma_j = 0.05 + \text{HalfNormal}(1)$ (the floor guards against a Heywood case); ordered-normal cutpoints τ_j ; the negative-binomial concentration $\alpha_j \sim \text{HalfNormal}(2)$; and $\Phi \sim \text{LKJ}(\eta=2)$, whose density $\propto \det(\Phi)^{\eta-1}$ gently favours weakly correlated factors [39]. Latent scores use a non-centred parameterization ($D_{ik} = z_{ik}$, $z_{ik} \sim \mathcal{N}(0, 1)$, scale carried by the loadings) to defuse the hierarchical funnel.

Why the prior is load-bearing: rotation. A factor model is not identified from the likelihood alone. For any invertible M ,

$$\Lambda^* = \Lambda M, \quad \Phi^* = M^{-1} \Phi M^{-\top}, \quad f^* = M^{-1} f \implies \Lambda^* \Phi^* \Lambda^{*\top} + \Psi = \Lambda \Phi \Lambda^\top + \Psi = \Sigma, \quad (13)$$

so rotated parameters imply the identical Σ , hence the identical Gaussian likelihood, and are indistinguishable by any amount of data. Confirmatory factor analysis breaks this by fixing loadings to exactly zero; the soft-prior ESEM breaks it with the prior. The likelihood is rotation-invariant but the prior is *not*: a rotation moves a loading the prior wanted near zero away from zero, lowering $p(\Lambda)$, so the posterior $p(\Lambda \mid X) \propto p(X \mid \Lambda) p(\Lambda)$ concentrates at the one orientation where small loadings align with the soft-zero priors and home loadings are positive and substantial; the positive truncation removes the residual sign-flip. A corollary gives the robustness check used below: under a flat loading prior the posterior mode equals the maximum-likelihood (FIML) solution, so a prior-free refit landing on the same loadings shows the structure is earned from the data, not manufactured by the priors.

4.6 The covariance identity and the headline estimand

For the Gaussian block the inner integral of (5) is closed-form, giving the identity at the heart of factor analysis. From (3), using $\text{Var}(G_i) = 1$, $\text{Cov}(D_i) = \Phi_D$, $\mathbb{E}[\varepsilon_i \varepsilon_i^\top] = \Psi$ diagonal, and the two independence facts that do the work— $\mathbb{E}[G_i D_i^\top] = \mathbf{0}$ (bifactor orthogonality) and $(G_i, D_i) \perp \varepsilon_i$ —the cross-terms in $\Sigma = \text{Cov}(x_i) = \mathbb{E}[(x_i - \mu_i)(x_i - \mu_i)^\top]$ vanish and the survivors sum to

$$\Sigma = \underbrace{\lambda_G \lambda_G^\top}_{\text{general: rank one}} + \underbrace{\Lambda_D \Phi_D \Lambda_D^\top}_{\text{specific factors}} + \underbrace{\Psi}_{\text{private noise}} = \Lambda \Phi \Lambda^\top + \Psi, \quad \Phi = \begin{pmatrix} 1 & \mathbf{0}^\top \\ \mathbf{0} & \Phi_D \end{pmatrix}, \quad (14)$$

the zero off-diagonal block being precisely the encoding of $G \perp$ specifics. The general factor leaves a *rank-one* footprint: every pair of items j, j' covaries through G by exactly $\lambda_{jG} \lambda_{j'G}$ —one shared source touching all items. Because x_i is a linear combination of jointly Gaussian quantities it is itself Gaussian, $x_i \sim \mathcal{N}(\mu_i, \Sigma)$.

The headline lives in the off-diagonal block. The correlated- G model *unzips* the zero block of (14), freeing $\phi_{Gd} = \text{Corr}(G, D)$:

$$\Phi = \begin{pmatrix} 1 & \phi_{Gd}^\top \\ \phi_{Gd} & \Phi_D \end{pmatrix}, \quad \Sigma = \lambda_G \lambda_G^\top + \underbrace{\lambda_G \phi_{Gd}^\top \Lambda_D^\top + \Lambda_D \phi_{Gd} \lambda_G^\top}_{\text{the terms orthogonality removed}} + \Lambda_D \Phi_D \Lambda_D^\top + \Psi. \quad (15)$$

The vector ϕ_{Gd} —the correlation of each specific factor with general burden—is the headline estimand of the enterprise: “biology is the least severity-entangled domain” is the statement that its metabolic and inflammatory entries are near zero (0.12 and 0.07) while cognition and sleep are far larger (0.39 and 0.42). The bifactor orthogonality is the *null*; the correlated- G arm measures the departure from it, and the bifactor direct loading $|\lambda_{jG}|$ and the correlation ϕ_{Gd} agree on the ordering—reporting both is what turns “biology is separable” from a constraint artefact into a finding.

4.7 Missing data, Woodbury computation and factor scores

Missing data is the marginal property. For patient i with observed continuous set $C_i \subseteq O_i$, a foundational property of the multivariate normal is that the marginal is obtained by *deleting* the unobserved coordinates from (μ, Σ) ,

$$X_{i,C_i} \sim \mathcal{N}(\mu_{i,C_i}, \Sigma_{C_i}), \quad \Sigma_{C_i} = \Lambda_{C_i} \Phi \Lambda_{C_i}^\top + \Psi_{C_i}, \quad (16)$$

with Λ_{C_i} the observed-row loadings and $\Psi_{C_i} = \text{diag}(\sigma_j^2)_{j \in C_i}$. “Integrating out” the missing cells is therefore not an extra step but the marginal property itself; summing $\log \mathcal{N}(\cdot)$ over each patient’s observed sub-vector is *exactly* the full-information maximum-likelihood (FIML) objective for incomplete multivariate-normal data. Hence the Bayesian marginalized model and FIML optimize the same observed-data likelihood and differ only by the priors (12); a standalone FIML estimator adds no independent evidence, and confirmation is done in-engine.

Woodbury evaluation at scale. Evaluating (16) naively costs $\mathcal{O}(|C_i|^3)$ per patient. The Woodbury identity and matrix-determinant lemma reduce it to the $(K+1)$ -dimensional factor space,

$$\Sigma_{C_i}^{-1} = \Psi_{C_i}^{-1} - \Psi_{C_i}^{-1} \Lambda_{C_i} (\Phi^{-1} + \Lambda_{C_i}^\top \Psi_{C_i}^{-1} \Lambda_{C_i})^{-1} \Lambda_{C_i}^\top \Psi_{C_i}^{-1}, \quad (17)$$

$$\log |\Sigma_{C_i}| = \log |\Phi^{-1} + \Lambda_{C_i}^\top \Psi_{C_i}^{-1} \Lambda_{C_i}| + \log |\Phi| + \sum_{j \in C_i} \log \sigma_j^2, \quad (18)$$

an $\mathcal{O}((K+1)^3)$ cost; patients sharing a missingness pattern share the inner $(K+1) \times (K+1)$ factorization, computed once per pattern. This removes the per-patient latent funnel for the high-dimensional continuous block and is what makes full-sample ($N = 9,013$) estimation tractable on a workstation (Apple M4 Pro, 24 GB, no GPU); the non-Gaussian block retains explicit per-patient latents.

Factor scores. Each patient’s coordinates are the expected a posteriori (EAP) factor scores from their observed cells; for an all-continuous pattern,

$$\hat{f}_i = \mathbb{E}[f_i | X_{i,C_i}] = \Phi \Lambda_{C_i}^\top \Sigma_{C_i}^{-1} (X_{i,C_i} - \mu_{i,C_i}), \quad S_i = \Phi - \Phi \Lambda_{C_i}^\top \Sigma_{C_i}^{-1} \Lambda_{C_i} \Phi, \quad (19)$$

obtained by MCMC when non-Gaussian cells are present. The per-patient posterior covariance S_i is exported and propagated downstream (the noise term in the stratification mixture, (21)): a patient with one observed indicator for a factor is *not* treated as equally characterized as one with six.

4.8 Staged estimation, priors and convergence

The global posterior was reached by continuation (homotopy): a well-conditioned sub-model was fit and then deformed toward the full target one source of difficulty at a time (S1 continuous core at full N ; S2 ESEM cross-loadings; S3 mixed-likelihood suicidality and developmental-risk; S4 thin cohort-specific factors; S5 all factors jointly plus the correlated- G variant), each converged posterior warm-starting the next. *Only the global S5 fit is interpreted*; the earlier stages are convergence/identification checkpoints (publishing an intermediate sub-model would be exactly the error to avoid). Sampling used the No-U-Turn sampler [40] via PyMC [41] with the NumPyro/JAX backend [42] on a workstation (Apple M4 Pro, 24 GB; no GPU). A stage advanced only on $\hat{R} \leq 1.01$, effective sample size ≥ 400 , zero divergences, no Heywood cases and acceptable posterior-predictive behaviour. The full- N continuous backbone met this strict gate; the joint nine-dimension (mixed) map did not certify at full N and is reported at the largest sample that mixes ($\hat{R} \leq 1.04$, effective sample size ≥ 112), with a cross-seed stability guard (Tucker congruence $\varphi = 0.993$).

4.9 In-engine confirmation

Three checks established that the structure is earned, not manufactured. (i) A prior-free refit with flat loading priors (identification constraints only) reproduced the loadings and Φ essentially exactly (Tucker $\varphi = 1.00$); a flat-prior MAP equals the MLE/FIML. (ii) Posterior-predictive checks compared model-implied and observed pairwise correlations, yielding a Bayesian SRMR ≈ 0.07 for the continuous block and reproduction of 21/22 non-Gaussian indicators’ endorsement rates/means within the 90% predictive interval (lone exception: a 90.8%-zero suicide-attempt count item, an item-level not factor-level misfit). (iii) Model comparison by WAIC/PSIS-LOO [43, 44] decisively preferred the bifactor over unidimensional and correlated-factor alternatives. Classical fit indices (CFI/TLI, RMSEA) are not reported—they are weak under heavy missingness and non-normal indicators and add no independent evidence here.

4.10 Invariance, robustness and patient scoring

Measurement invariance across BP/SZ/DR was tested by multi-group loadings and Bayesian differential item functioning; partials are documented, not hidden. Robustness to cohort imbalance was assessed by leave-one-cohort-out congruence, diagnosis-balanced subsampling, site cluster-bootstrap and $1/n_{\text{cohort}}$ weighting (minimum Tucker $\varphi = 0.958$); imbalance was handled by weighting and invariance testing rather than by discarding data. Each patient received, per axis, a posterior mean, SD and highest-density interval, the number of observed indicators, and

a reliability tier (well-characterized / partial / prior-dominated), so that downstream analyses propagate uncertainty rather than treating all coordinates as equally measured.

4.11 Stratification

The stratification operates on the fitted coordinates $\{x_i, S_i\}_{i=1}^N$ with $x_i = \hat{f}_i \in \mathbb{R}^D$ ($D = 9$) and known posterior covariance S_i from (19); uncertainty is carried throughout rather than discarded.

Structure-discovery gate (is there cluster structure at all?). Before any clustering we test the cloud’s shape with five complementary statistics, each computed over posterior draws so that measurement noise cannot manufacture structure. *Cluster tendency* uses the Hopkins statistic

$$H = \frac{\sum_{i=1}^m u_i^D}{\sum_{i=1}^m u_i^D + \sum_{i=1}^m w_i^D}, \quad (20)$$

where w_i is the nearest-neighbour distance of a sampled real point and u_i that of a uniform point over the data support; $H \approx \frac{1}{2}$ indicates no clustering. *Modality* uses Hartigan’s dip statistic [45] $\text{dip}(F_n) = \min_{U \in \mathcal{U}} \sup_x |F_n(x) - U(x)|$ ($\mathcal{U}$ the unimodal CDFs) on PC1 and each axis. *Model selection* uses the gap statistic [46], $\text{Gap}(K) = \mathbb{E}^*[\log W_K] - \log W_K$ with within-cluster dispersion $W_K = \sum_k \frac{1}{2|C_k|} \sum_{i,i' \in C_k} \|x_i - x_{i'}\|^2$, alongside the Gaussian-mixture BIC profile. *Density* uses HDBSCAN [47] (fraction of points assigned to noise), and *topology* a Mapper graph [48]. A continuum verdict ($H \rightarrow \frac{1}{2}$, unimodal dip, Gap maximized at $K = 1$, flat BIC basin, HDBSCAN all-noise, a single connected Mapper component) makes archetypes the lead view; a clustered verdict would promote the mixture.

Model A — measurement-error Gaussian mixture. Treating x_i as a noisy reading of a latent position, $x_i \mid \theta_i \sim \mathcal{N}(\theta_i, S_i)$, with θ_i drawn from a K -component mixture yields the extreme-deconvolution likelihood [49]

$$x_i \sim \sum_{k=1}^K \pi_k \mathcal{N}(m_k, \Sigma_k + S_i), \quad \pi \sim \text{Dirichlet}\left(\frac{\alpha}{K} \mathbf{1}\right), \quad (21)$$

in which *every* observation carries its own additive, known covariance S_i , so imprecise coordinates spread their mass over components (diffuse responsibility) and prior-dominated or cohort-absent axes self-down-weight. The soft decision regions are the posterior responsibilities

$$r_{ik} = \frac{\pi_k \mathcal{N}(x_i \mid m_k, \Sigma_k + S_i)}{\sum_{l=1}^K \pi_l \mathcal{N}(x_i \mid m_l, \Sigma_l + S_i)}, \quad (22)$$

with optional hard label $\arg \max_k r_{ik}$ and assignment-confidence $\max_k r_{ik}$. A small Dirichlet concentration ($\alpha \ll 1$) makes (21) a sparse/overfitted finite mixture, so the effective number of regions is data-selected; K is triangulated by BIC, PSIS-LOO [44] and cross-seed stability, and label switching is resolved by a relabeling algorithm.

Model B — archetypal analysis (lead view under a continuum). Each patient is represented as a convex combination of A extreme phenotypes, each itself a convex combination of patients [50]:

$$\min_{W,B} \sum_{i=1}^N \left\| x_i - \sum_{a=1}^A w_{ia} z_a \right\|^2 \quad \text{s.t.} \quad z_a = \sum_{j=1}^N b_{ja} x_j, \quad w_{ia}, b_{ja} \geq 0, \quad \sum_a w_{ia} = 1, \quad \sum_j b_{ja} = 1, \quad (23)$$

i.e. $X \approx WZ$ with $Z = BX$ and both W and B row/column-stochastic, which forces the archetypes z_a onto the convex hull of the cloud. The simplex weights $w_i \in \Delta^{A-1}$ are the soft

archetype membership (a patient as “0.7 high-inflammatory + 0.3 cognitive”). Uncertainty is propagated by refitting (23) across posterior draws and by projecting each draw onto the fixed archetypes (simplex-constrained least squares), giving a posterior over w_i ; the number of archetypes A is set by the explained-variance scree, cross-resample stability (Tucker congruence), and interpretability, and—under a continuum—is a parsimony choice rather than a natural number.

Validation. The partition/archetypes are tested on four gates. Variance explained per axis, $\eta_d^2 = 1 - \sum_k \sum_{i \in \mathcal{C}_k} (x_{id} - \bar{x}_{kd})^2 / \sum_i (x_{id} - \bar{x}_d)^2$, quantifies whether structure is driven by the specific/biological axes rather than G (not-just-severity). Concordance with diagnosis uses the adjusted Rand index $\text{ARI} = (\text{RI} - \mathbb{E}[\text{RI}]) / (\max \text{RI} - \mathbb{E}[\text{RI}])$ and Cramér’s $V = \sqrt{\chi^2 / (N \min(r - 1, c - 1))}$ against cohort and the seven DSM-5 subtypes (transdiagnosticity). Stability is the cross-seed ARI/Tucker congruence, and a missingness-artefact check compares the accuracy of a classifier predicting membership from the per-axis coverage pattern against the majority baseline. A free mixture is compared with one constrained to the DSM-5 partition by information criteria (the “tighter description than diagnosis” test). UMAP [51] and the principal-component embedding (Results Fig. 4) are used for visualization only, never as clustering inputs.

4.12 Temporal coherence

Follow-up visits were scored on the fixed measurement model (observed cells, uncertainty propagated). Longitudinal metric invariance was tested by per-visit loading congruence vs baseline. Trait/state was decomposed with a measurement-error random-intercept model per axis, plugging the known baseline measurement variance (so a low-reliability axis cannot masquerade as state) and including visit fixed effects (so a shared population trajectory is not miscounted as individual instability), yielding a population “slide” and an individual rank-stability intraclass correlation. A complementary per-patient geometric analysis counted a coordinate move only when it cleared measurement error, and quantified archetype-identity persistence (κ vs the chance rate). Attrition was characterized by logistic retention~baseline coordinates; inverse-probability-of-retention weights were available for sensitivity analysis.

4.13 Prognosis

For each two-year outcome we compared nested models—nuisance + baseline outcome + severity (the reference), then adding DSM-5 subtype, the map, or both—using an errors-in-variables Bayesian generalized linear model that propagated each coordinate’s posterior SD. Incremental value was the held-out expected log-predictive density and, for binary endpoints, the change in area under the ROC curve with decision-curve analysis. Robustness used inverse-probability weighting for attrition, restriction to reliably measured axes, and a permutation null. “Predicts” denotes incremental association under internal validity, not causation; endpoints are state transitions from repeated functioning/severity scales, not registered events.

4.14 Treatment

Treatment exposures (recovered from per-cohort source dictionaries and harmonized to common drug classes) were analysed by a causal pipeline: a common-support overlap gate on the propensity $P(\text{treat} \mid \text{severity} + \text{diagnosis} + \text{demographics} + \text{the nine coordinates})$ [52]; covariate balance before/after stabilized inverse-probability-of-treatment weighting [53]; and a doubly-robust errors-in-variables moderation model with the durable-axis $\times$ treatment interaction as the estimand. Average treatment effects carried an E-value sensitivity analysis for unmeasured

confounding [54]. A treatment-as-confounder analysis re-fit the functioning prognosis with and without the drug-class exposures.

4.15 Reproducibility

The pipeline is configuration-first with fixed seeds; raw per-cohort data are confidential and never leave their governed environment, while code, the harmonized dictionary and regenerated aggregate results are shareable. Every reported number is reproducible from the numbered pipeline (build → fit → confirm → invariance → score → robustness → stratify → temporal → prognosis → treatment); Table 1 is regenerated by an aggregate-only script.

Data availability

The FACE database is the property of Fondation FondaMental; per-patient cohort data are confidential and governed by FondaMental’s data-access procedures, to which qualified researchers may apply. No per-patient data are reproduced in this article. Aggregate parameters, derived structure and all figure/table source values are available in the project’s regenerated aggregate outputs.

Code availability

The full analysis pipeline (data harmonization, the Bayesian measurement engine, stratification, temporal, prognosis and treatment modules), the harmonized variable dictionary, and configuration files are available at [TODO: repository URL / DOI]. Every reported quantity is reproducible from the numbered pipeline; per-patient data are never required to regenerate the aggregate results, tables and figures.

Ethics

The FACE cohorts were approved by the relevant institutional review boards and all participants provided written informed consent. [TODO: Insert exact IRB/ethics committee names, approval numbers, and trial/cohort registration identifiers.]

Author contributions

[TODO: Complete per ICMJE / journal policy and FACE–FondaMental authorship conventions, including cohort investigators.]

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